

- 1 A cuboid has a square area $(6 - \sqrt{35}) \text{ m}^2$. The capacity of the cuboid is $(45\sqrt{5} - 38\sqrt{7}) \text{ m}^3$. Find the height of the cuboid, in m, in the form $(a\sqrt{5} - b\sqrt{7}) \text{ m}$, where a and b are integers. [3]

- 2 **(a)** Show that $(x - 1)$ is a factor of $f(x) = x^3 - (2h + 1)x^2 - (k - 2h)x + k$,
where h and k are constants. [1]
- (b)** Explain with working why the equation $f(x) = 0$ will have 3 real roots if
 $h^2 + k \geq 0$. [4]

3 The temperature, T °C, of glass being heated in a kiln is given by the formula

$T = 2030 - 2000e^{kt}$, where t is the time in hours since the glass was placed in the kiln. The temperature of the glass after 90 minutes is 1000 °C.

(a) Find the temperature of the glass before it was heated. [1]

(b) The glass can be molded into different shapes when its temperature reaches at least 1300 °C. What is the earliest time that the glass can be removed from the kiln and molded? [4]

(c) The instruction manual of the kiln gives the range of the temperature of the kiln as $T < y$ °C. State the value of y . [1]

- 4 The curve $y = f(x)$ is such that $f''(x) = \frac{18}{(1-2x)^3}$ and $2f'(-1) = 1$. Find the equation of the curve given that it passes through the point $(1, 0)$. [6]

- 5** It is given that $f(x) = \frac{(x+2)^2}{p-x}$, $x \neq p$, where p is an integer.
- (a) Obtain and simplify an expression for $f'(x)$ in terms of p . [3]
- (b) The range of values of x for which $f(x)$ is an increasing function is $-2 < x < 8$. Show that the value of p is 3. [3]
- (c) Hence find the equation of the normal to the curve at the point where the curve crosses the x -axis. [3]

- 6** It is given that $5 \sin^2 A - 3 \cos^2 A = 7 \sin 2A$ where $0^\circ \leq A \leq 90^\circ$.
- (a)** Show that $\tan A = 3$. [3]
- (b)** Hence find the value of $\cos(60^\circ + A)$, leaving your answer in the form $\frac{a + b\sqrt{3}}{2\sqrt{10}}$, where a and b are integers. [3]
- (c)** Without finding the value of A , explain whether $60^\circ + A$ is acute or obtuse. [1]

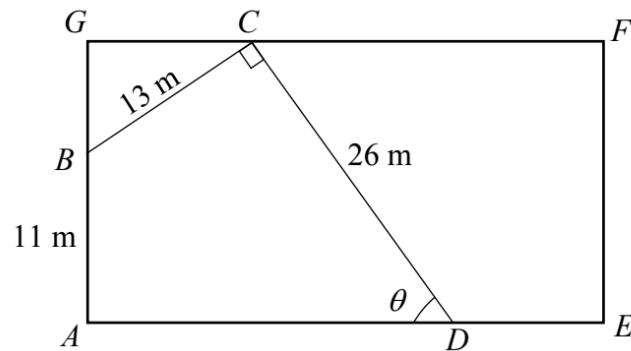
7 **(a)** Given that $y = (3x - 1)\sqrt{6x + 1}$, show that $\frac{dy}{dx} = \frac{27x}{\sqrt{6x + 1}}$. [3]

(b) Hence find the value of $\int_0^4 \frac{9x - 3}{\sqrt{6x + 1}} dx$. [4]

- 8** **(a)** In the binomial expansion of $\left(2x - \frac{3}{x}\right)^n$, where n is a positive integer, the coefficient of the third term is $\frac{270}{8}(2^n)$. Show that $n = 6$. [4]
- (b)** Using the value of n in **part (a)**, find the coefficient of x^4 in the expansion of $\left(1 + x^2\right)\left(2x - \frac{3}{x}\right)^n$. [3]

- 9 It is given that $y = a \cos 2x + b$ for $-180^\circ \leq x \leq 90^\circ$ where $a < 0$ and $b > 0$. It is given that the amplitude of y is 3 and the maximum value of y is 7.
- (a) State the value of a and of b . [2]
- (b) Sketch the graph of $y = a \cos 2x + b$ for $-180^\circ \leq x \leq 90^\circ$. [3]
- (c) Hence state the number of solution(s) of the equation $1 - \frac{1}{2} \cos 2x = 0$ for $-180^\circ \leq x \leq 90^\circ$. [2]

- 10** **(a)** Solve $49^x - 7^{x+1} = 18$, leaving your answer in the form $\log_a b$, where a and b are integers to be determined. [5]
- (b)** Find the largest value of the integer q such that $-3x^2 + qx - 8$ is negative for all real values of x . [3]



The diagram shows a rectangular futsal court, $AEFG$.

From a point A on the court, players are to run along the straight paths AB , BC , CD and DA . The lengths of AB , BC and CD are 11 m, 13 m and 26 m respectively. Angle ADC is θ , where $0^\circ < \theta < 90^\circ$. The total distance covered by each player is T metres.

- (a) Show that T can be expressed as $50 + 26 \cos \theta + 13 \sin \theta$. [3]
- (b) Express T in the form $50 + R \cos (\theta - \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. [3]
- (c) Bryan claims that he ran a total distance of 85 m. Explain with calculations if this is possible. [2]

- 12** The line $y = 4$ and the positive y -axis are tangents to a circle C . It is given that the x -coordinate of the centre of C is a , where $a > 0$.
- (a) Write down, in terms of a , the largest possible y -coordinate of the centre of C . [1]
The line L is a tangent to C at $(8, 13)$ on the circle. The centre of C lies below and to the left of $(8, 13)$.
- (b) Hence find the equation of C . [4]
- (c) Find the equation of L . [2]

- 13** **(a)** Express $\frac{4x^3 + 2x^2 - 5}{x^2(2x - 1)}$ in partial fractions. [7]
- (b)** Hence find $\int \frac{4x^3 + 2x^2 - 5}{x^2(2x - 1)} dx$. [3]